# FOUR EQUIVALENT DESCRIPTIONS OF A RELATIVELY VERY AMPLE INVERTIBLE SHEAF FOR A QUASI-COMPACT MORPHISM

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-relatively-very-ample is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 9145-9169 and proof span 9171-9237 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The author writes 'we urge the reader to prove this for themselves' at the point where immersiveness has to be shown, and later 'We omit the verification.' after describing the factorization of the associated morphism through the open subscheme. Both sentences are kept verbatim; the counted span nevertheless contains the reduction, the construction of the open subscheme U_j and the factorization, so what is left to the reader is the verification that the displayed factorization exhibits an immersion, not the mechanism of the argument.

*Editorial note.* 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- A relatively very ample sheaf makes the morphism separated, cited from the same chapter, together with locality of separatedness on the base. Cited at proof lines 9181-9183. Gives separatedness needed for the pushforward of the sheaf to be quasi-coherent and for the formulation of the fourth condition.
- Pushforward along a quasi-compact separated morphism preserves quasi-coherence, cited from the schemes chapter. Cited at proof lines 9169-9171. Makes the pushforward of the invertible sheaf a quasi-coherent module so that its projective bundle and the associated morphism make sense.
- The description of a morphism into a projective bundle by a surjection from a pullback of a module, and the adjunction with a map into the pushforward, cited from the constructions chapter. Cited at proof lines 9186-9190. Converts the local immersions into projective bundles into the surjectivity statement of the fourth condition.

*Date*: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The open subscheme of morphisms of relative Proj's induced by a map of modules, and the closed immersion of relative Proj's induced by a surjection from a symmetric algebra generated in degree one. Cited at proof lines 9200-9203 and 9229-9233. Supplies the open U_j through which the associated morphism factors, and the final implication from the third condition to the first.

*Source reference.* Original lemma 07ZG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10146–10146. Same-chapter reference used in the retained author text at source line(s) 9208. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 07ZK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10150–10150. Same-chapter reference used in the retained author text at source line(s) 9074, 9235, 10528. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original section 01OA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2187–2187. Same-chapter reference used in the retained author text at source line(s) 9189. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01VQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2455–2455. Same-chapter reference used in the retained author text at source line(s) 9182. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2081–2081. Same-chapter reference used in the retained author text at source line(s) 262, 404, 882, 1765, 1924, 2256, 2296, 8726, 9175, 10731, 13903. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible sheaf on $X$. Assume $f$ is quasi-compact. The following are equivalent*

1. *$\mathcal{L}$ is relatively very ample on $X/S$,*
2. *there exists an open covering $S = \bigcup V_j$ such that $\mathcal{L}|_{f^{-1}(V_j)}$ is relatively very ample on $f^{-1}(V_j)/V_j$ for all $j$,*
3. *there exists a quasi-coherent sheaf of graded $\mathcal{O}_S$-algebras $\mathcal{A}$ generated in degree 1 over $\mathcal{O}_S$ and a map of graded $\mathcal{O}_X$-algebras $\psi : f^*\mathcal{A} \to \bigoplus_{n\geq 0} \mathcal{L}^{\otimes n}$ such that $f^*\mathcal{A}_1 \to \mathcal{L}$ is surjective and the associated morphism $r_{\mathcal{L},\psi} : X \to \underline{Proj}_S(\mathcal{A})$ is an immersion, and*
4. *$f$ is quasi-separated, the canonical map $\psi : f^*f_*\mathcal{L} \to \mathcal{L}$ is surjective, and the associated map $r_{\mathcal{L},\psi} : X \to \mathbf{P}(f_*\mathcal{L})$ is an immersion.*

*Proof.* It is clear that (1) implies (2). It is also clear that (4) implies (1); the hypothesis of quasi-separation in (4) is used to guarantee that $f_*\mathcal{L}$ is quasi-coherent via Schemes, Lemma 01LC.

Assume (2). We will prove (4). Let $S = \bigcup V_j$ be an open covering as in (2). Set $X_j = f^{-1}(V_j)$ and $f_j : X_j \to V_j$ the restriction of $f$. We see that $f$ is separated by Lemma 01VQ (as being separated is local on the base). By assumption there exists a quasi-coherent $\mathcal{O}_{V_j}$-module $\mathcal{E}_j$ and an immersion $i_j : X_j \to \mathbf{P}(\mathcal{E}_j)$ with $\mathcal{L}|_{X_j} \cong i_j^*\mathcal{O}_{\mathbf{P}(\mathcal{E}_j)}(1)$. The morphism $i_j$ corresponds to a surjection $f_j^*\mathcal{E}_j \to \mathcal{L}|_{X_j}$, see Constructions, Section 01OA. This map is adjoint to a map $\mathcal{E}_j \to f_*\mathcal{L}|_{V_j}$ such that the composition

$$f_j^*\mathcal{E}_j \to (f^*f_*\mathcal{L})|_{X_j} \to \mathcal{L}|_{X_j}$$

is surjective. We conclude that $\psi : f^*f_*\mathcal{L} \to \mathcal{L}$ is surjective. Let $r_{\mathcal{L},\psi} : X \to \mathbf{P}(f_*\mathcal{L})$ be the associated morphism. We still have to show that $r_{\mathcal{L},\psi}$ is an immersion; we urge the reader to prove this for themselves. The $\mathcal{O}_{V_j}$-module map $\mathcal{E}_j \to f_*\mathcal{L}|_{V_j}$ determines a homomorphism on symmetric algebras, which in turn defines a morphism

$$\mathbf{P}(f_*\mathcal{L}|_{V_j}) \supset U_j \longrightarrow \mathbf{P}(\mathcal{E}_j)$$

where $U_j$ is the open subscheme of Constructions, Lemma 07ZG. The compatibility of $\psi$ with $\mathcal{E}_j \to f_*\mathcal{L}|_{V_j}$ shows that $r_{\mathcal{L},\psi}(X_j) \subset U_j$ and that there is a factorization

$$X_j \xrightarrow{r_{\mathcal{L},\psi}} U_j \longrightarrow \mathbf{P}(\mathcal{E}_j)$$

We omit the verification. This shows that $r_{\mathcal{L},\psi}$ is an immersion.

At this point we see that (1), (2) and (4) are equivalent. Clearly (4) implies (3). Assume (3). We will prove (1). Let $\mathcal{A}$ be a quasi-coherent sheaf of graded $\mathcal{O}_S$-algebras generated in degree 1 over $\mathcal{O}_S$. Consider the map of graded $\mathcal{O}_S$-algebras $\mathrm{Sym}(\mathcal{A}_1) \to \mathcal{A}$. This is surjective by hypothesis and hence induces a closed immersion

$$\underline{\mathrm{Proj}}_S(\mathcal{A}) \longrightarrow \mathbf{P}(\mathcal{A}_1)$$

which pulls back $\mathcal{O}(1)$ to $\mathcal{O}(1)$, see Constructions, Lemma 07ZK. Hence it is clear that (3) implies (1). □